

# **FEATURES OF THE SKULL**

**Dr. Osman Ahmed Osman**  
**Department of Anatomy**



# Learning Objectives

- Identify the individual bones which form the skull.
- Identify all of the bones of the cranium and the facial bones.
- Describe the different parts of each individual bone.

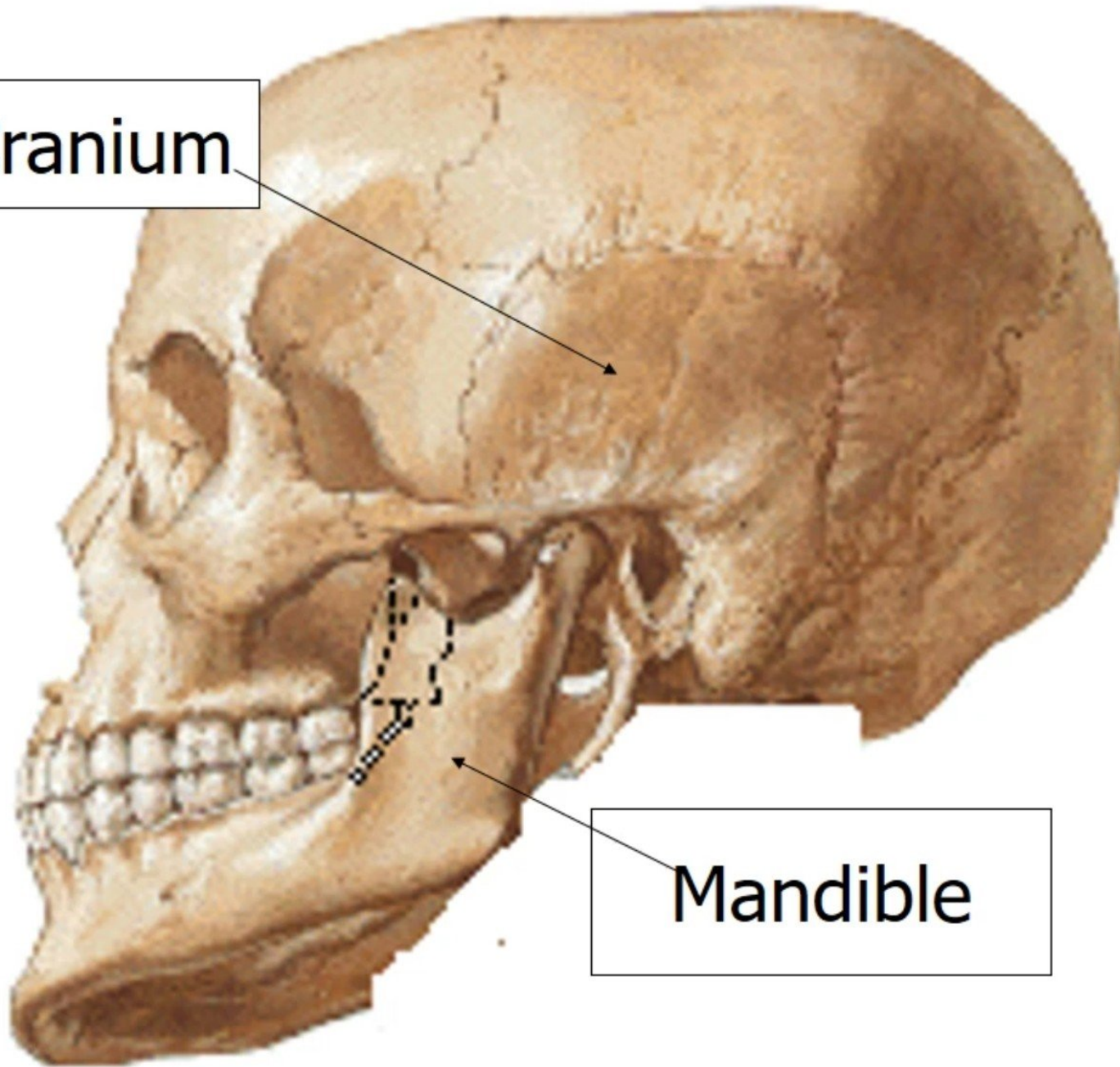


# SKULL

- The human skull is a bony structure, part of the skeleton, that is in the human head and which supports the structures of the face and forms a cavity for the brain.
- In humans, the adult **skull** is normally made up of 22 bones. Except for the mandible, all of the bones of the skull are joined together by sutures, (immovable) joints formed by bony ossification.
- The skull also contains the sinus cavities, which are air-filled cavities lined with respiratory epithelium, which also lines the large airways.



Cranium



Mandible





Pericranium

Dipploe

Endocranium



# FRAMEWORK OF THE SKULL

## Single Bones:-

- Frontal
- Occipital
- Sphenoid
- Ethmoid
- Vomer
- Mandible

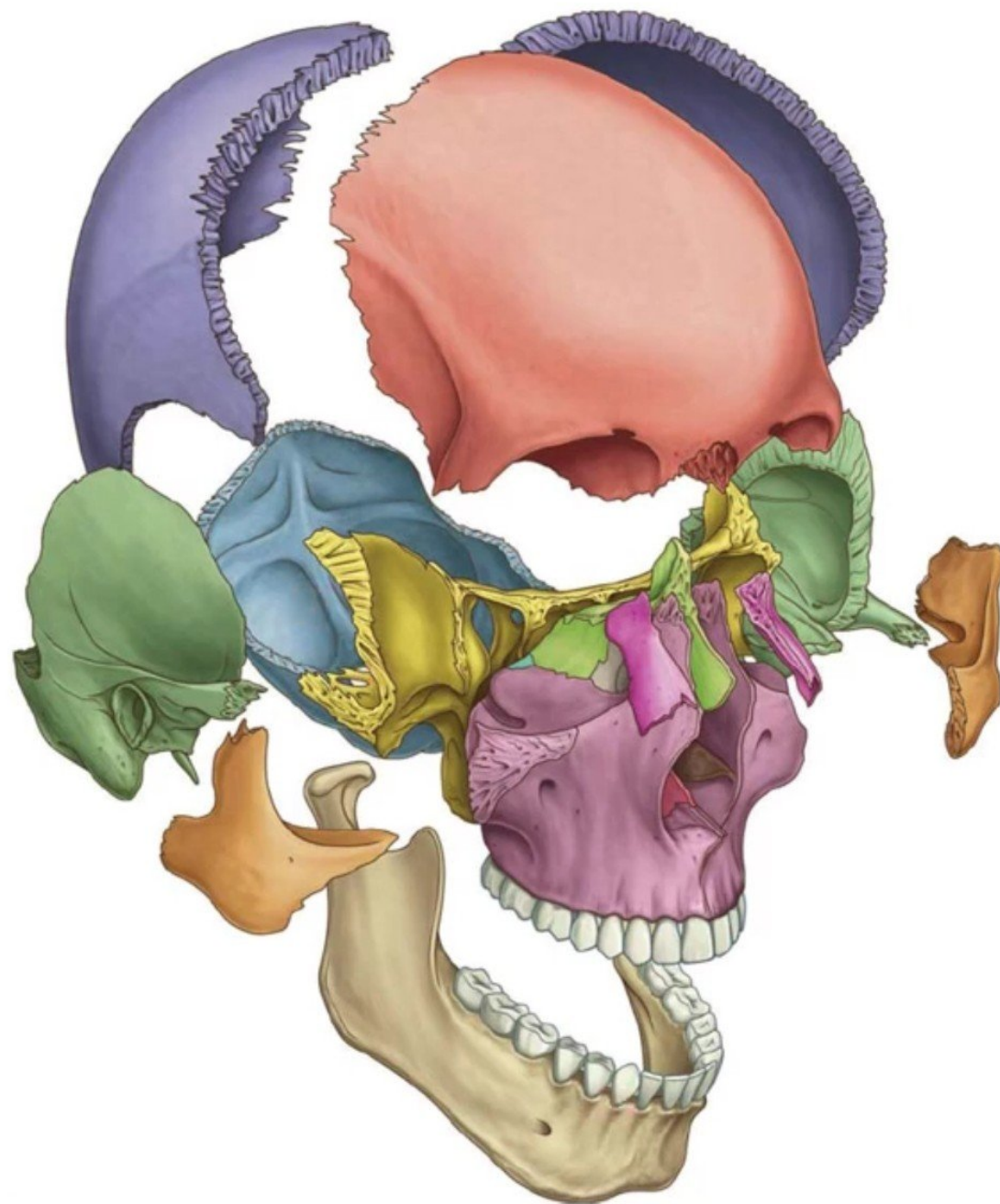


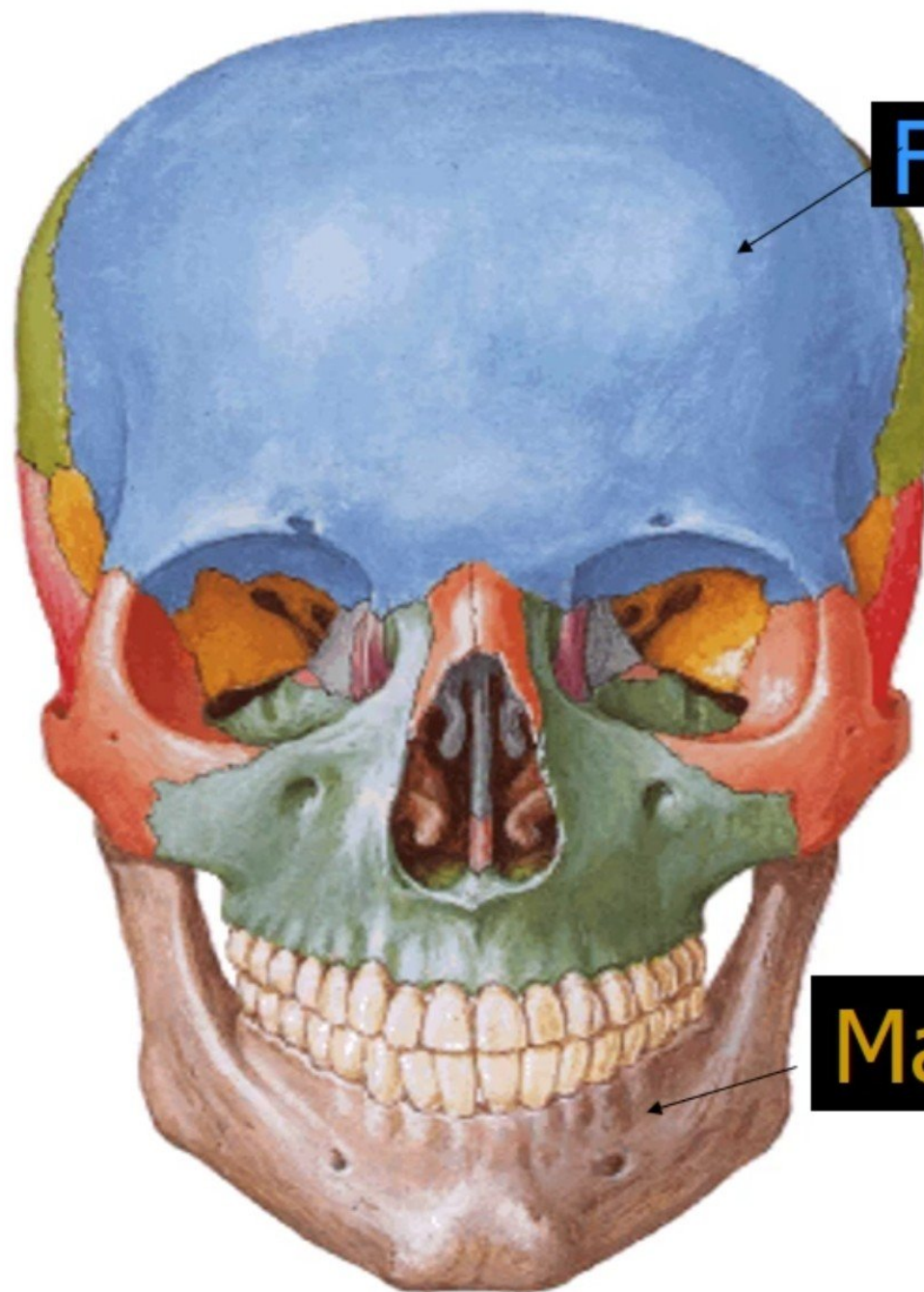
# FRAMEWORK OF THE SKULL

## Piers Bones:-

- Parietal
- Temporal
- Zygomatic
- Nasal
- Lacrimal
- Maxillae
- Inferior conchae
- Palatine



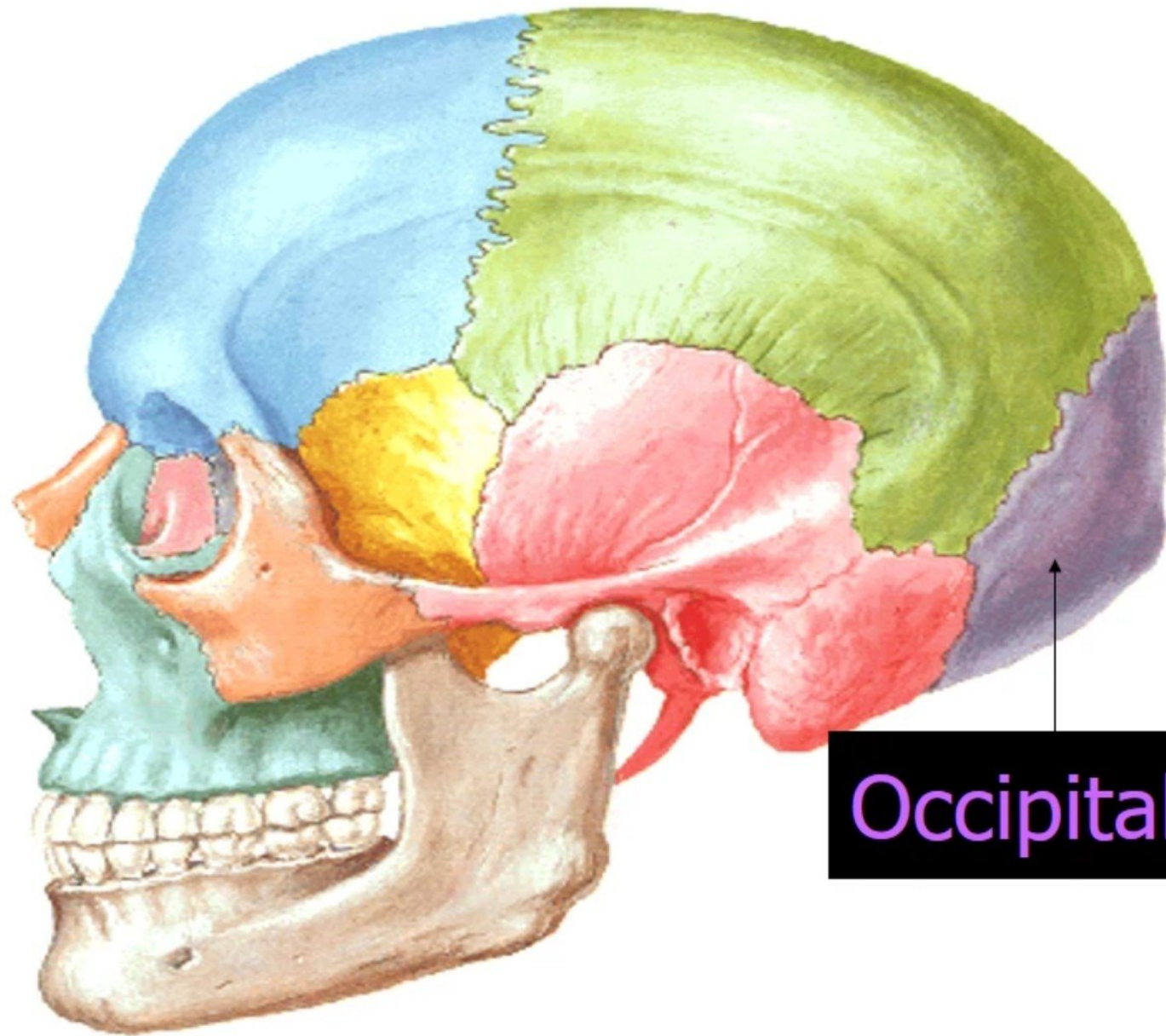




Frontal

Mandible

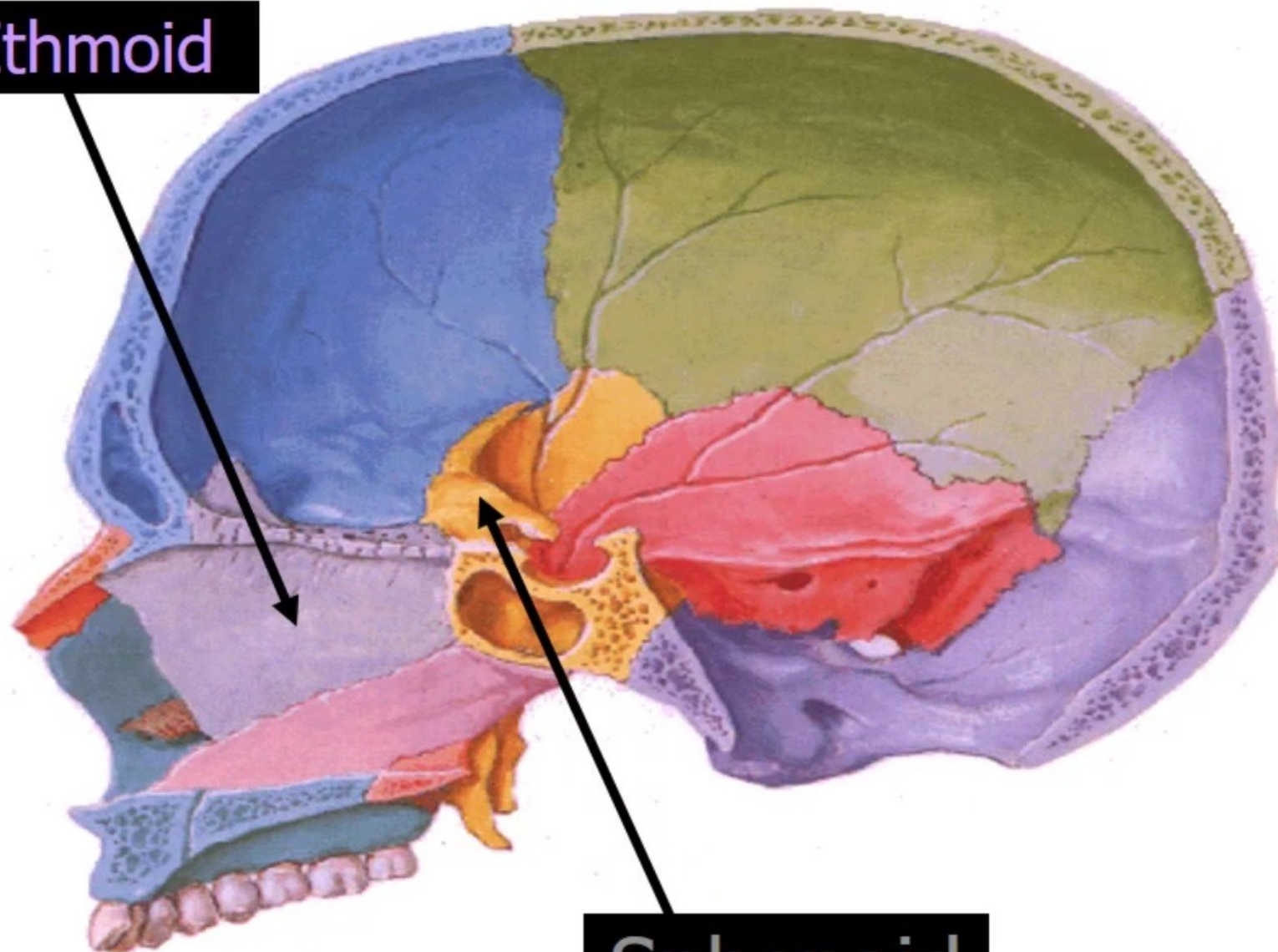




Occipital

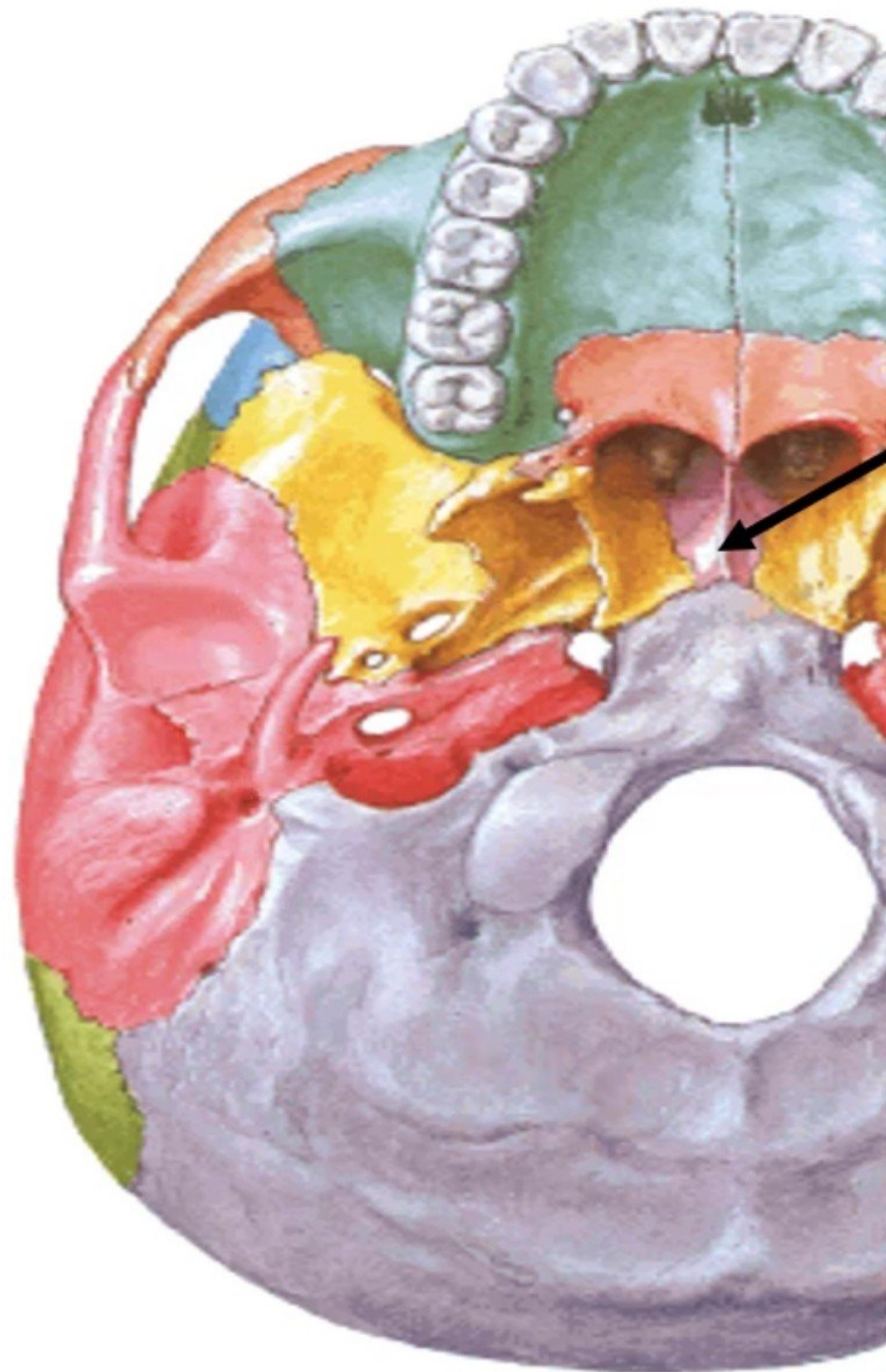


Ethmoid



Sphenoid





vomer

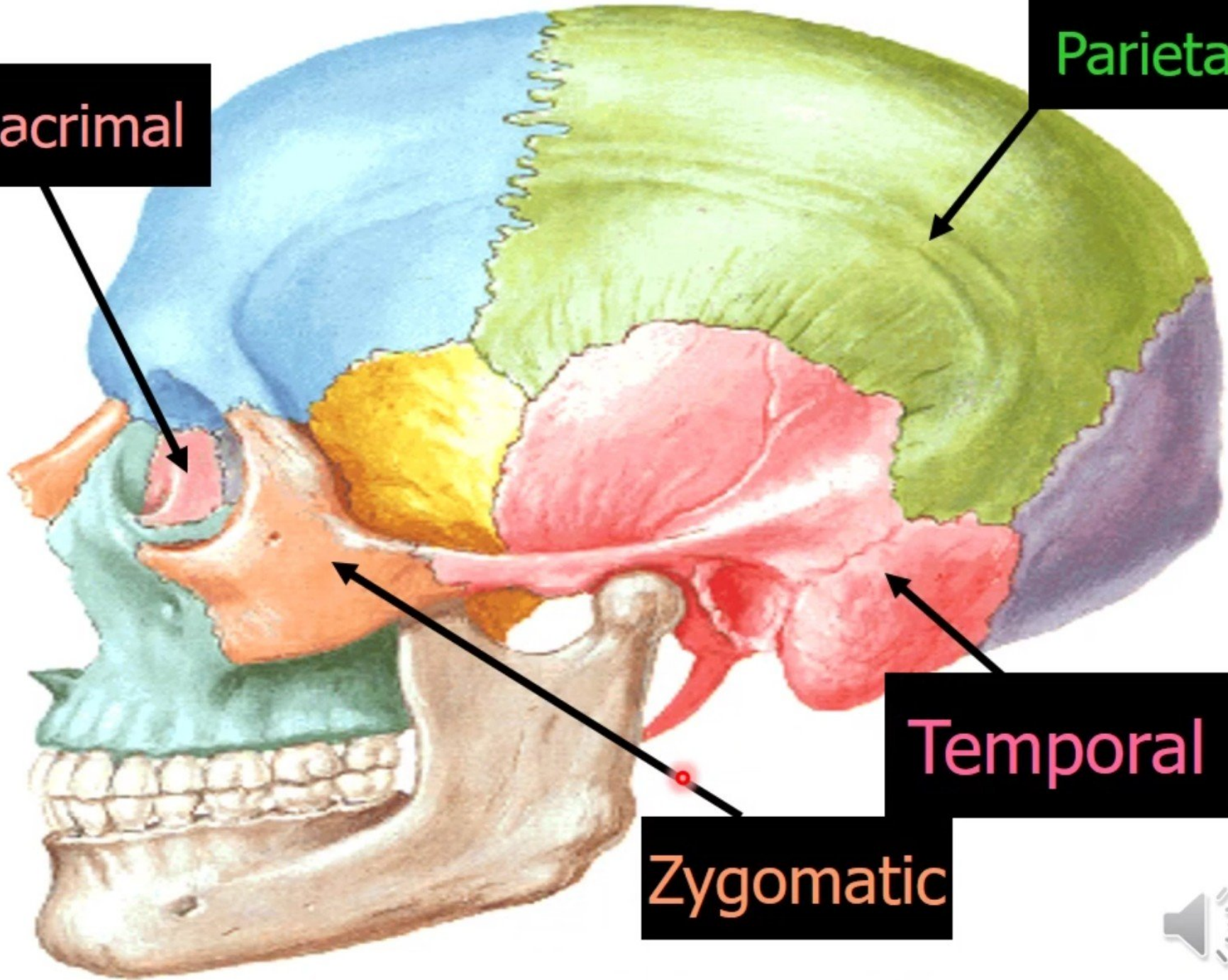


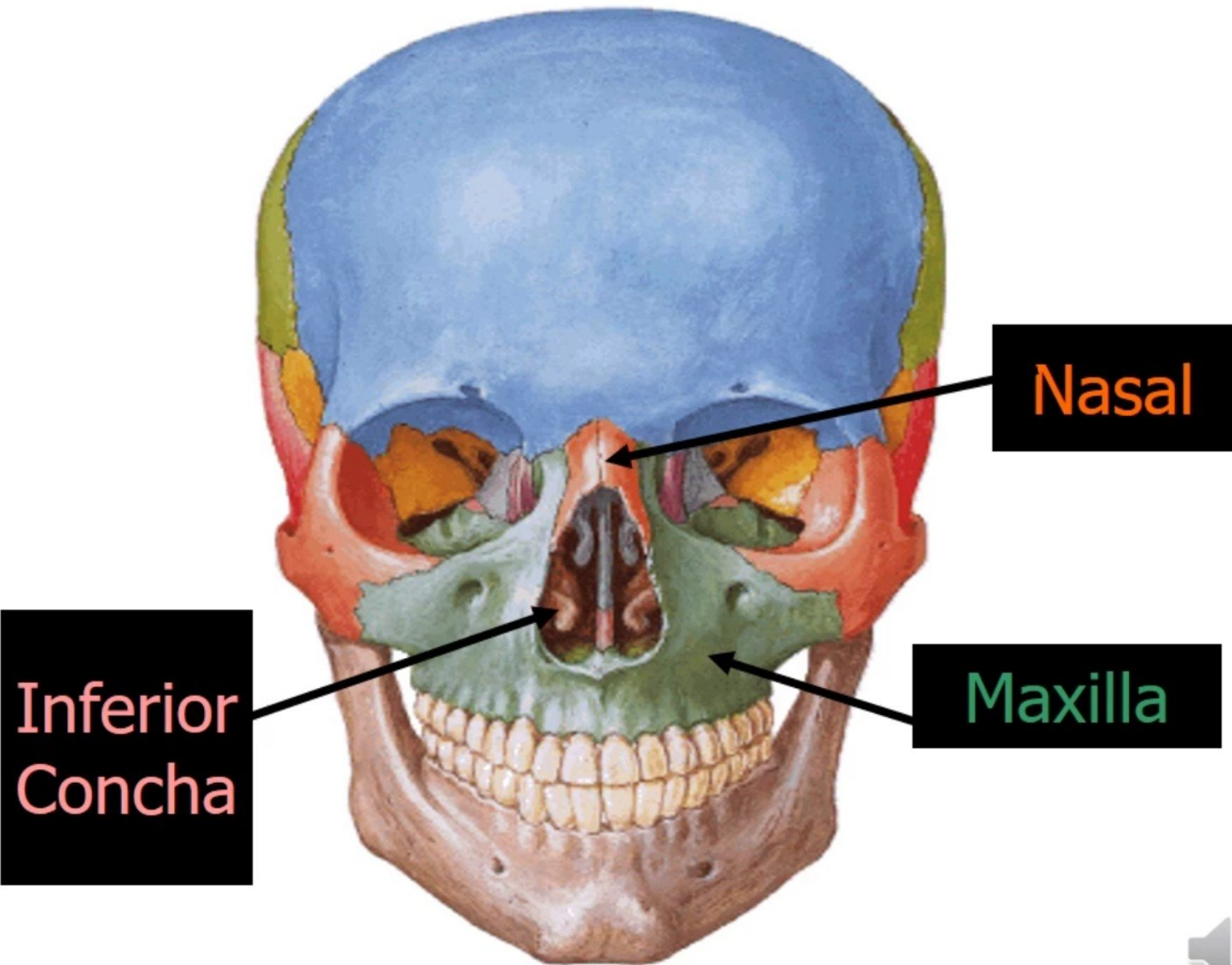
Lacrimal

Parietal

Temporal

Zygomatic







Palatine

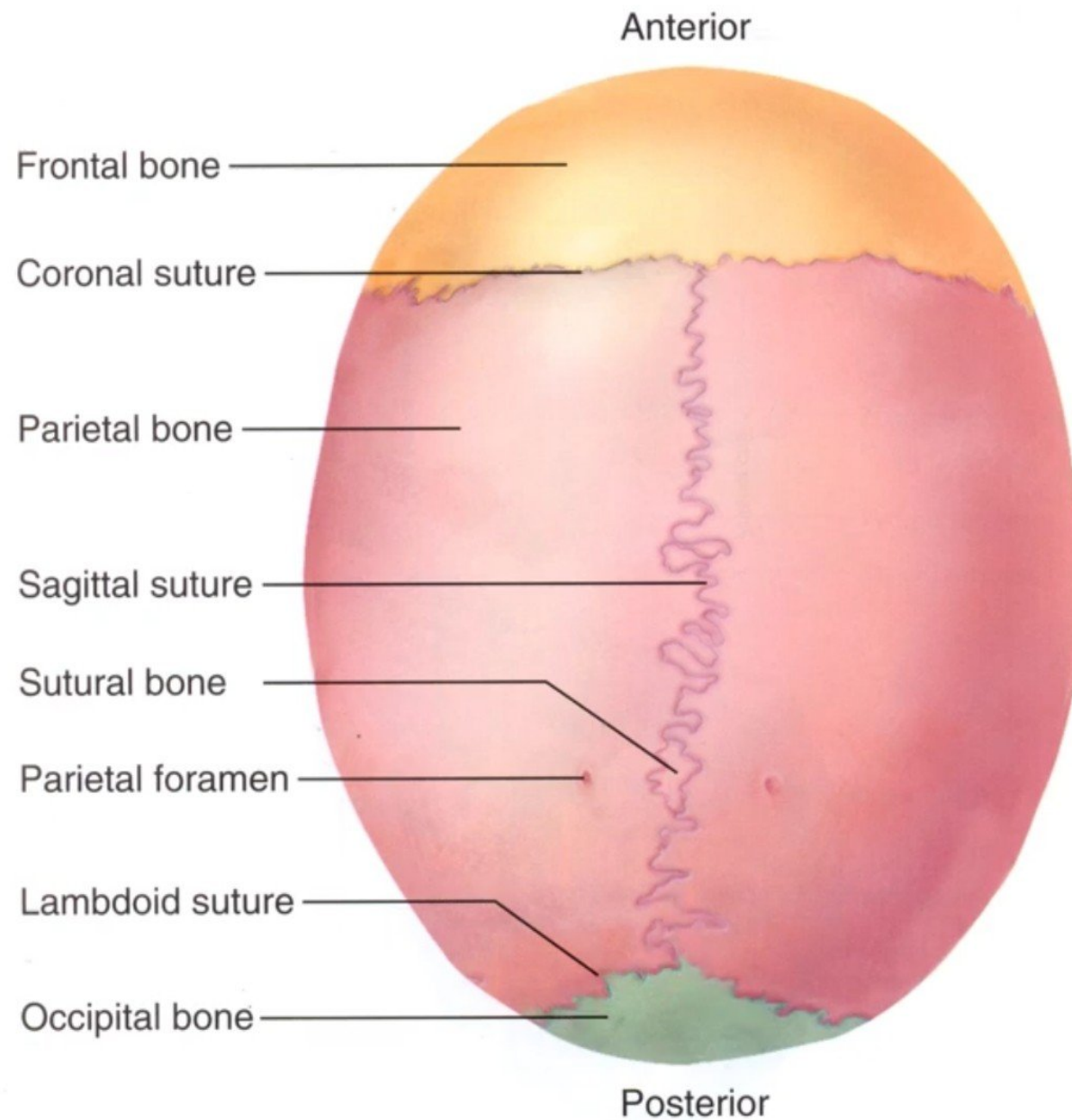


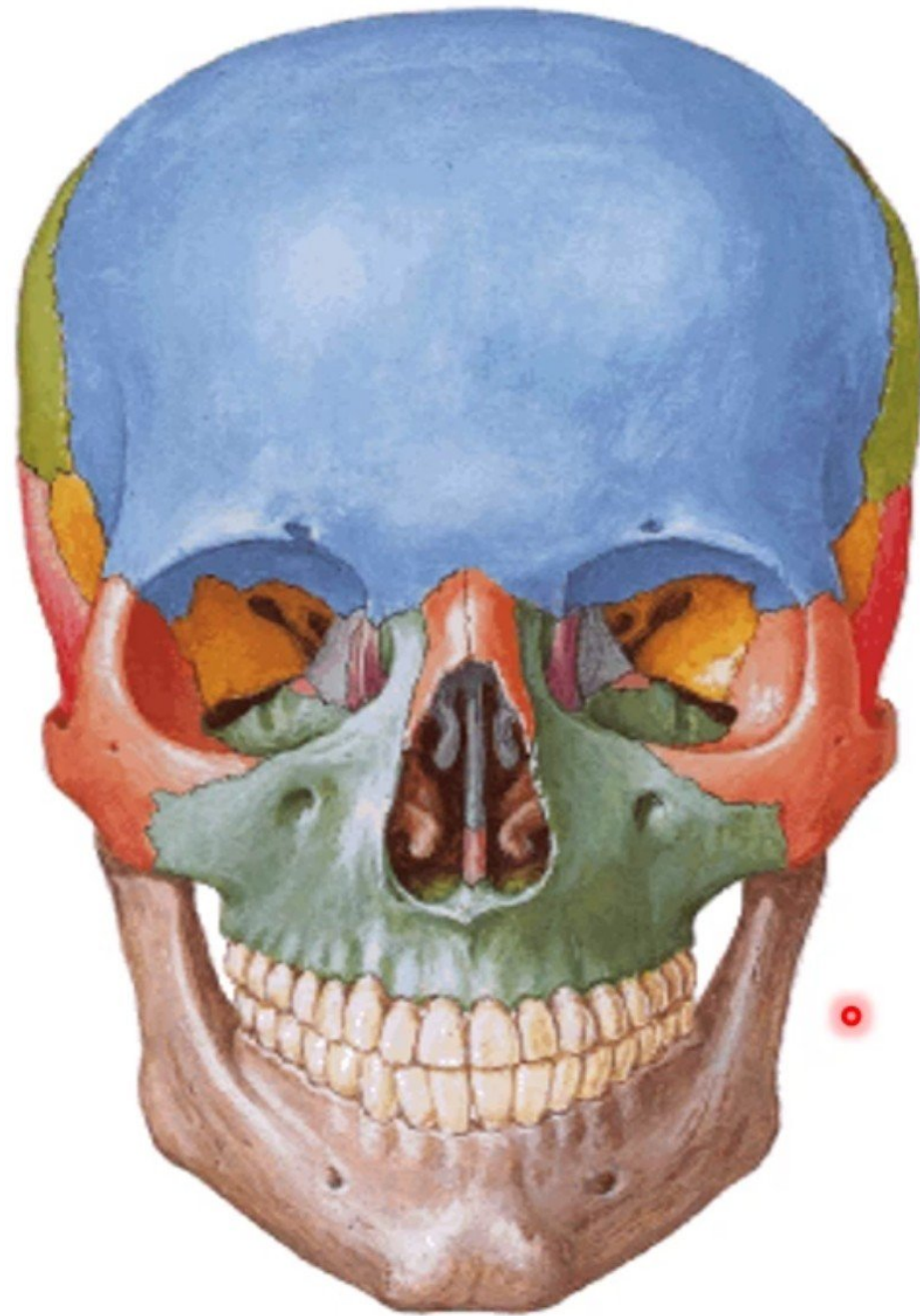
# **NORMAS OF THE SKULL**

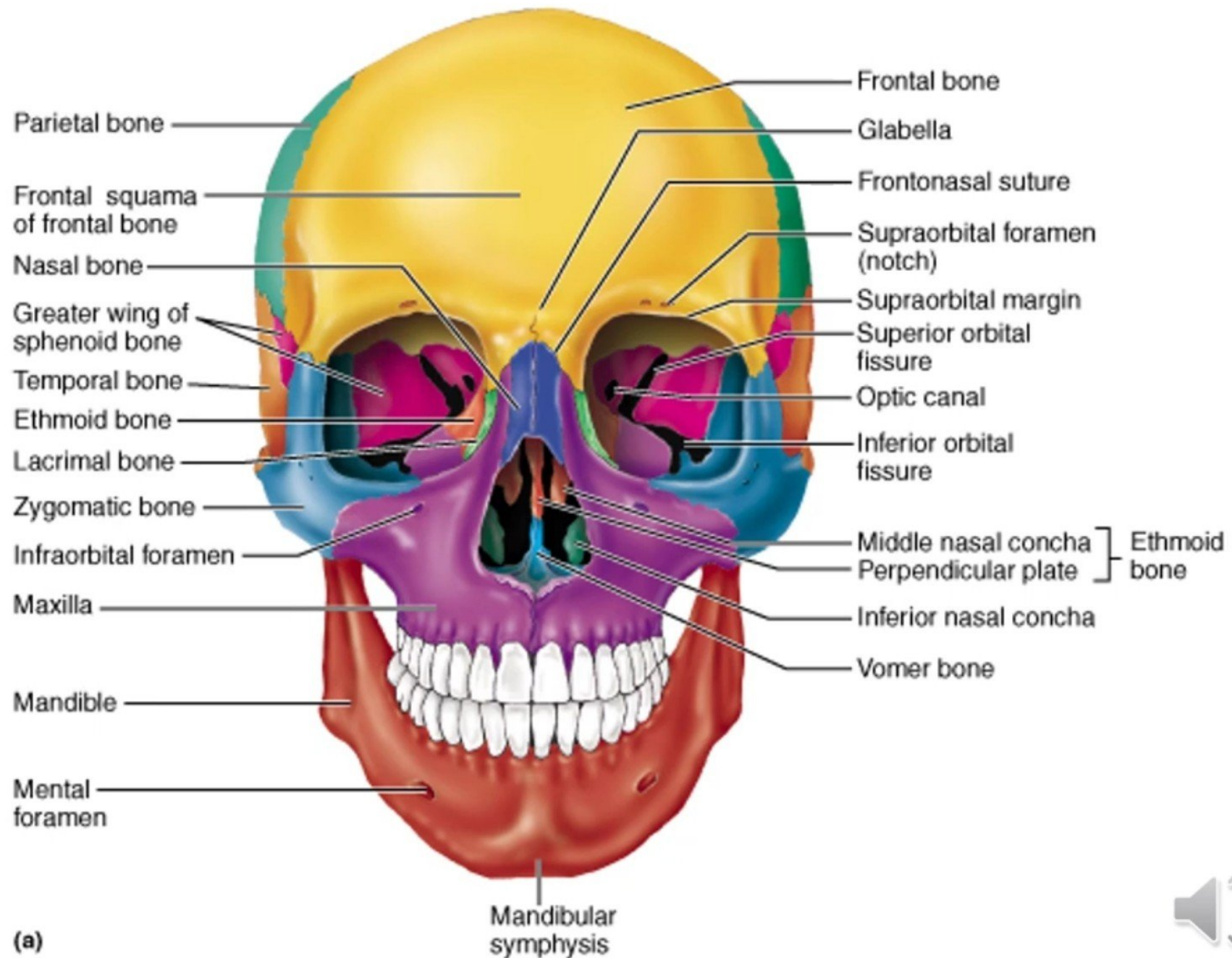
- Norma verticalis
- Norma frontalis
- Norma lateralis
- Norma occipitalis
- Norma basalis externa

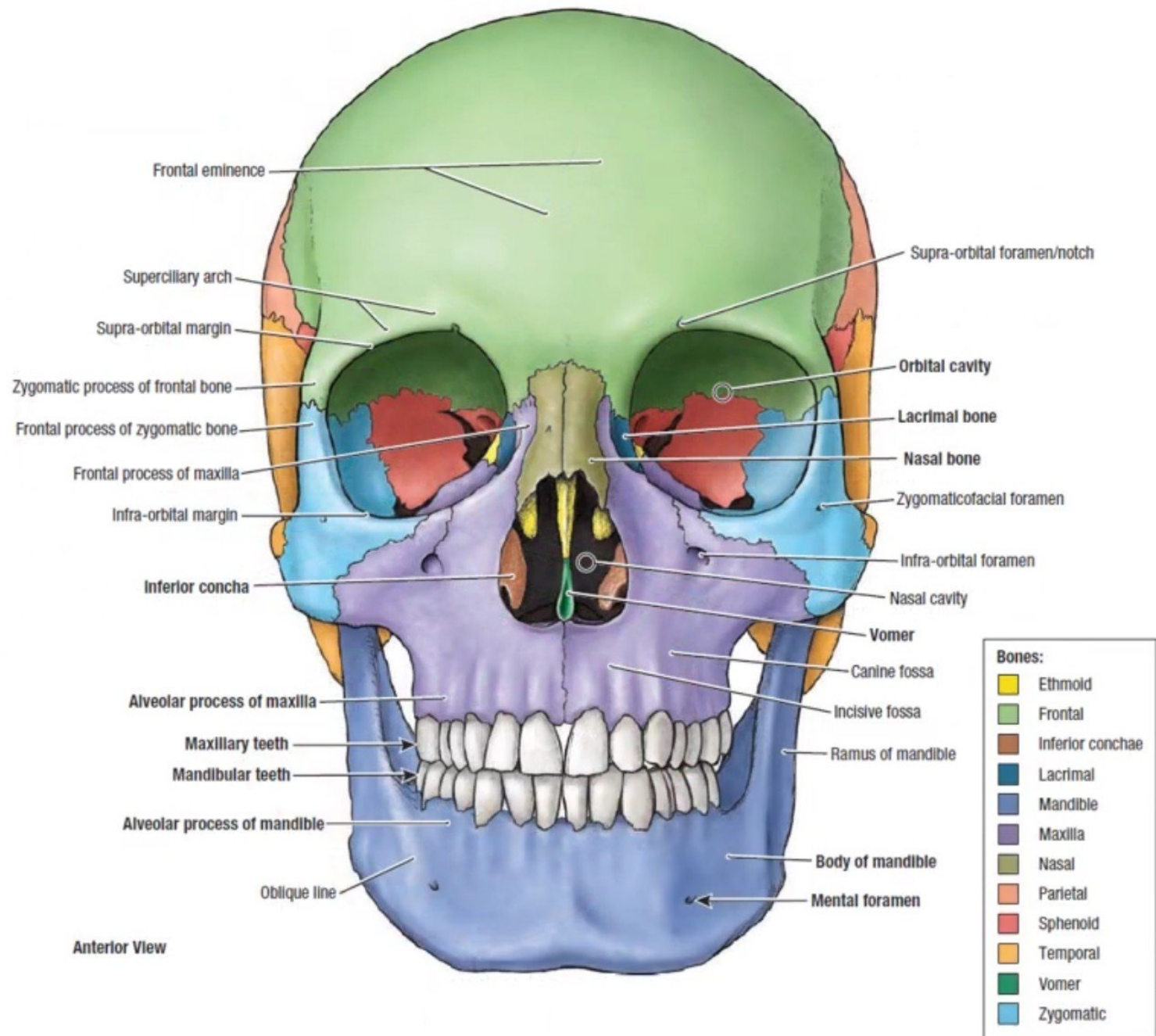


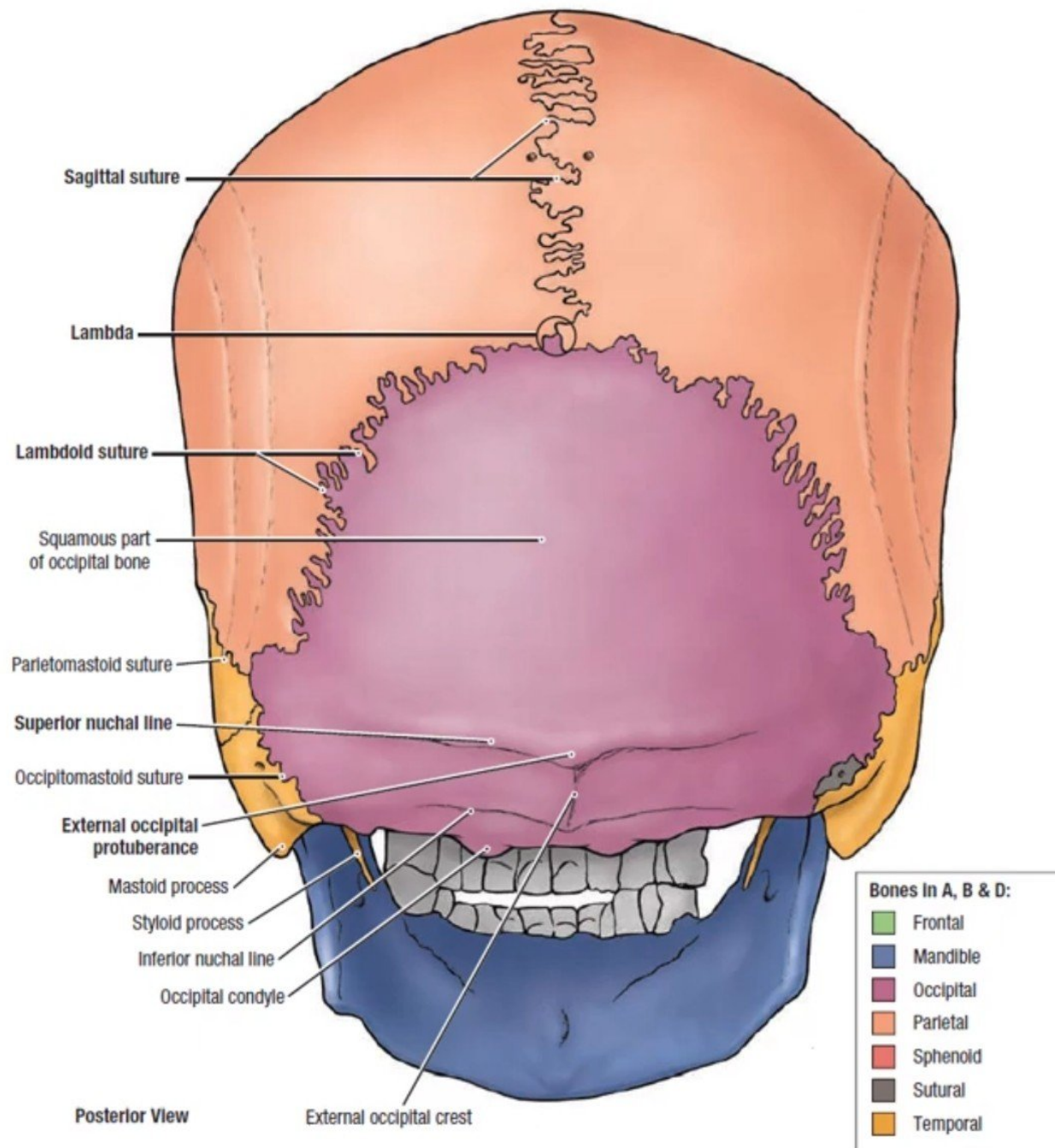


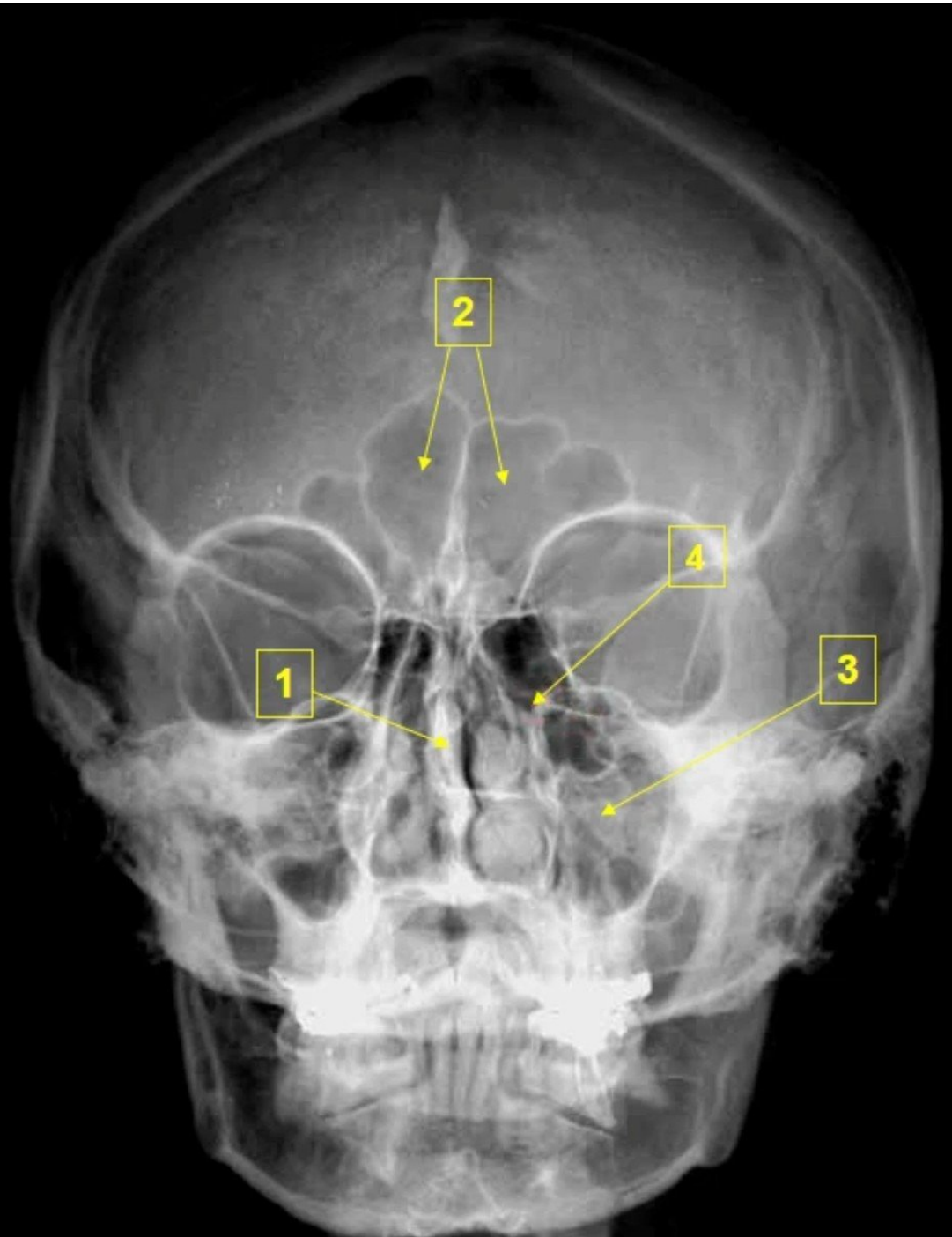








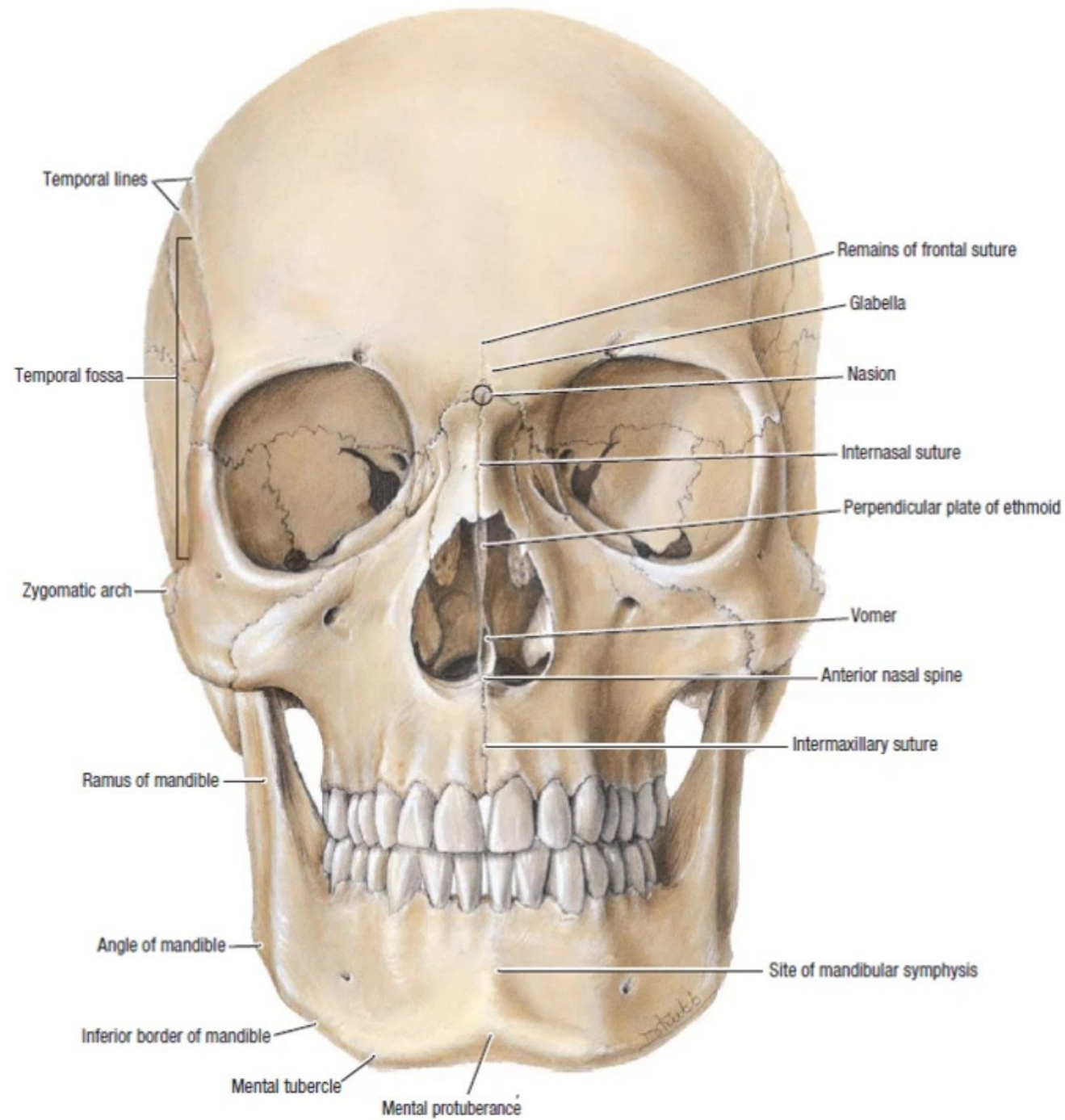


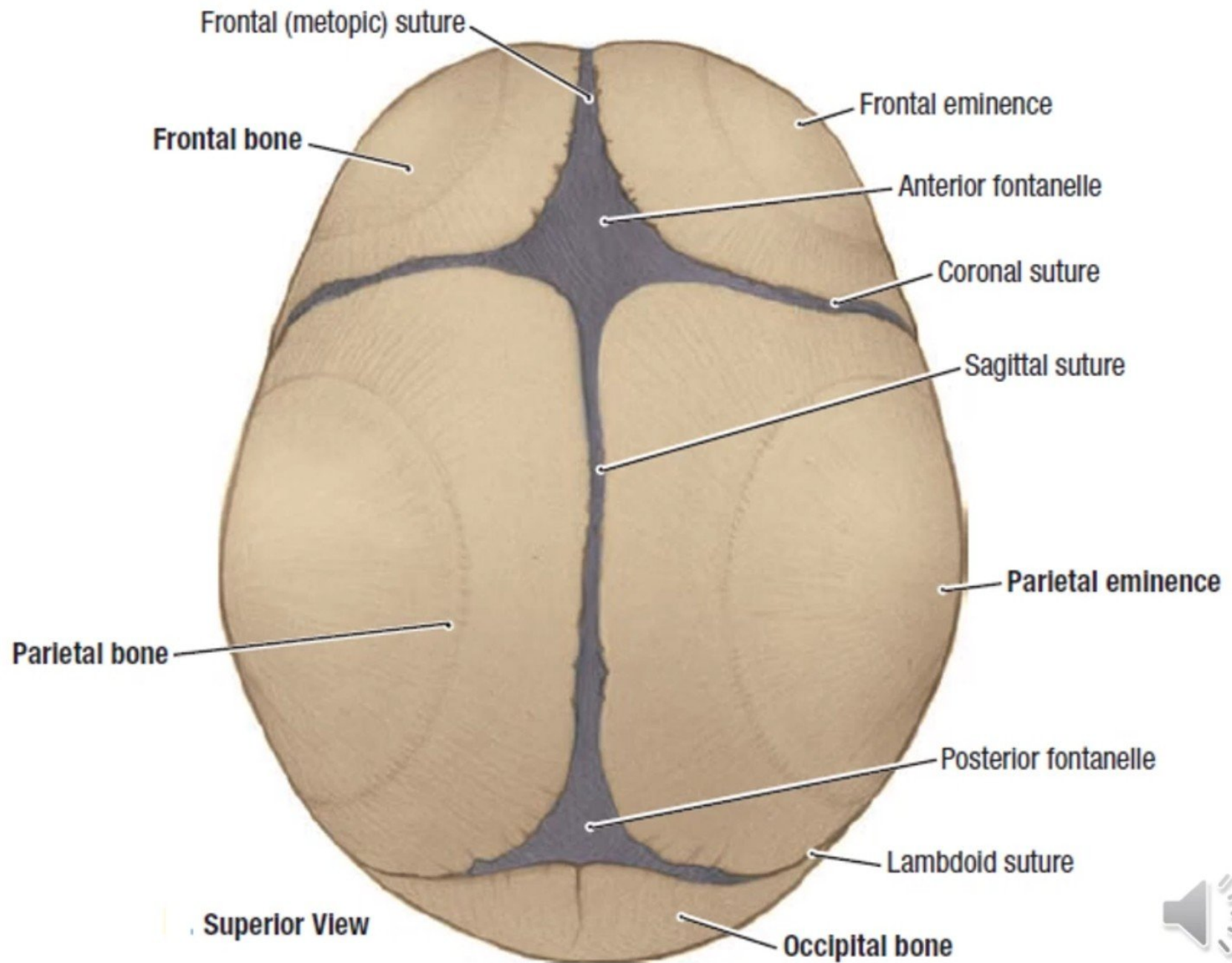


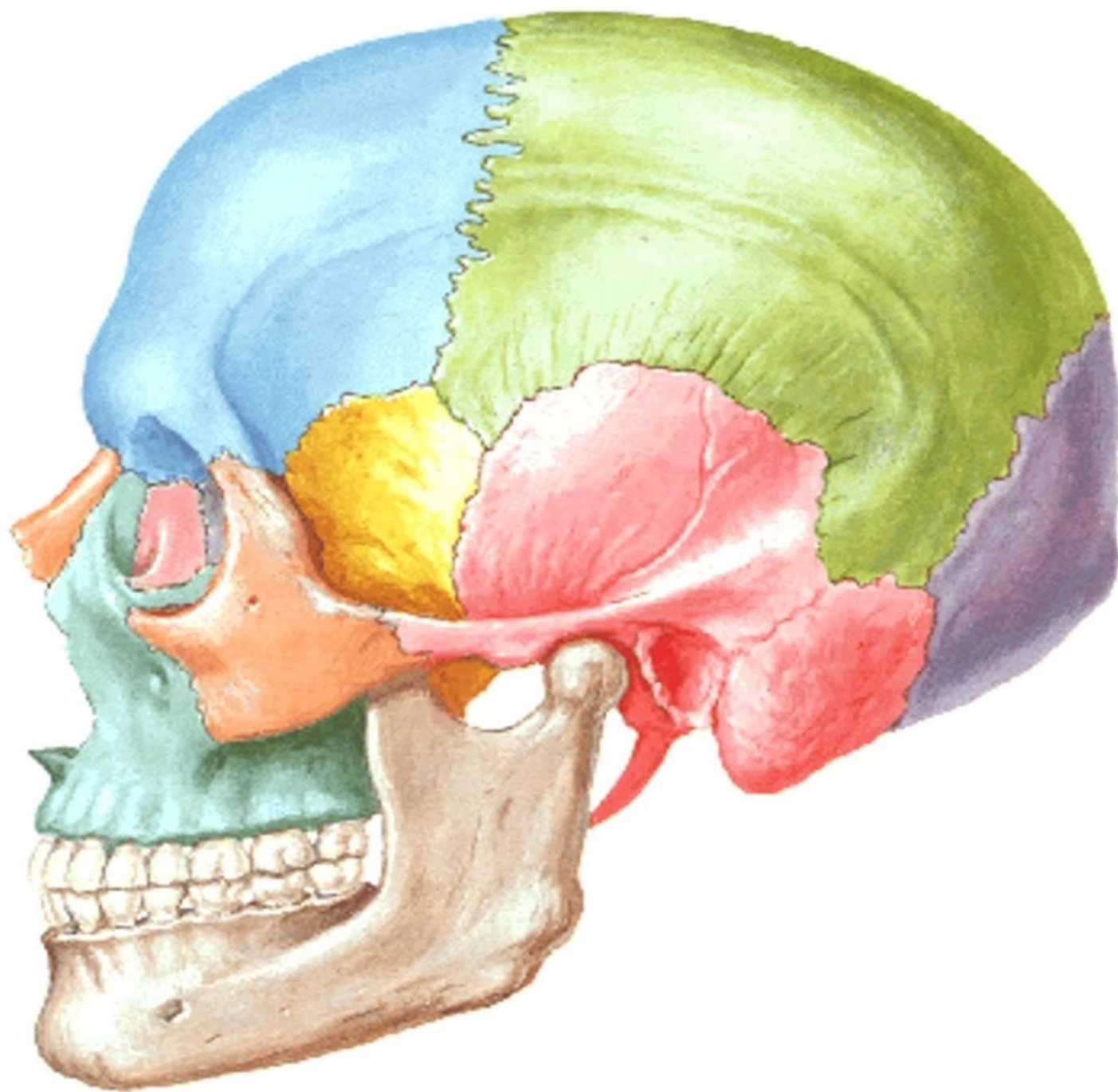
## AP View of Sinuses

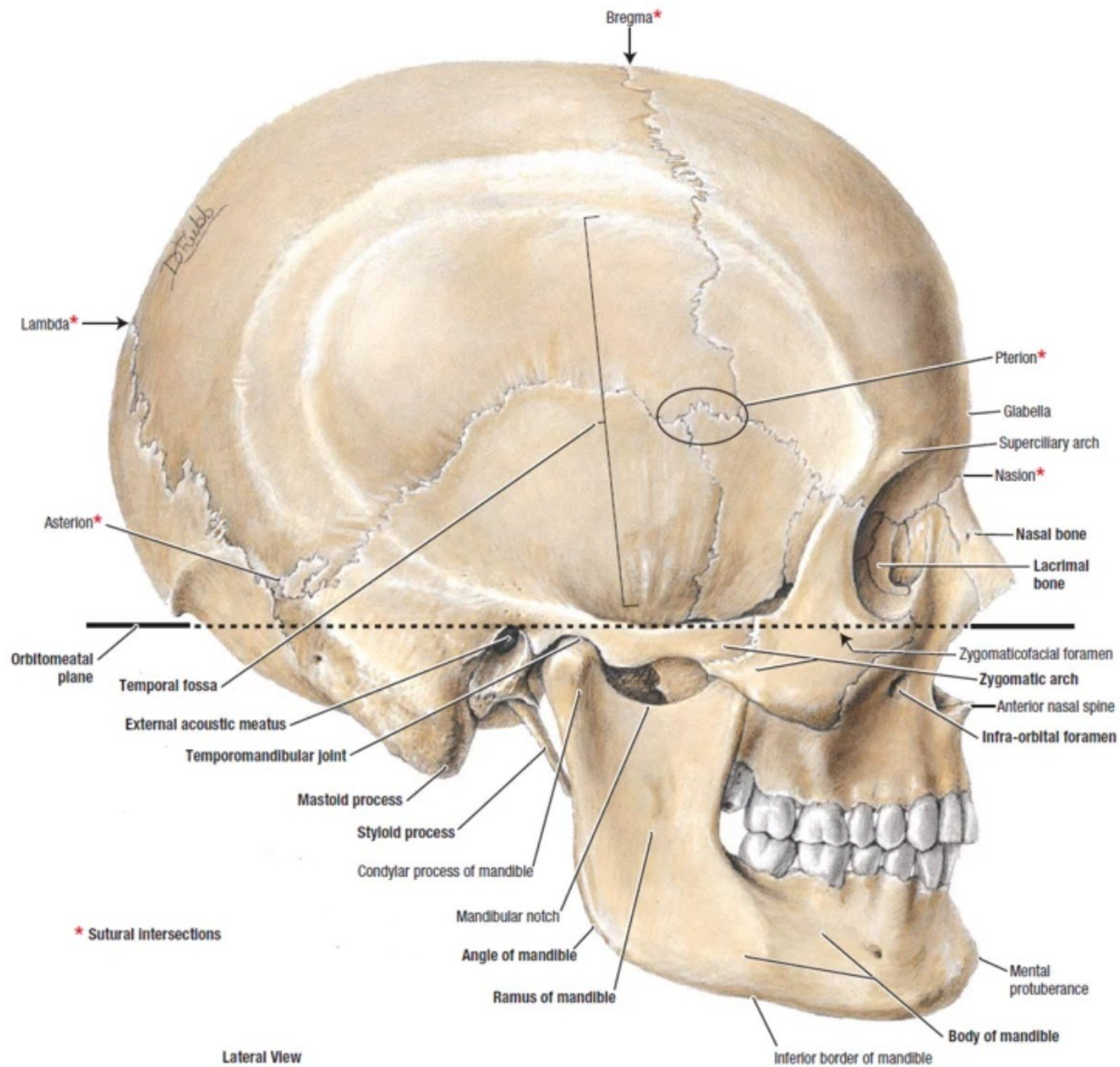
1. Nasal Septum
2. Frontal Sinus
3. Maxillary Sinus
4. Ethmoid Sinus





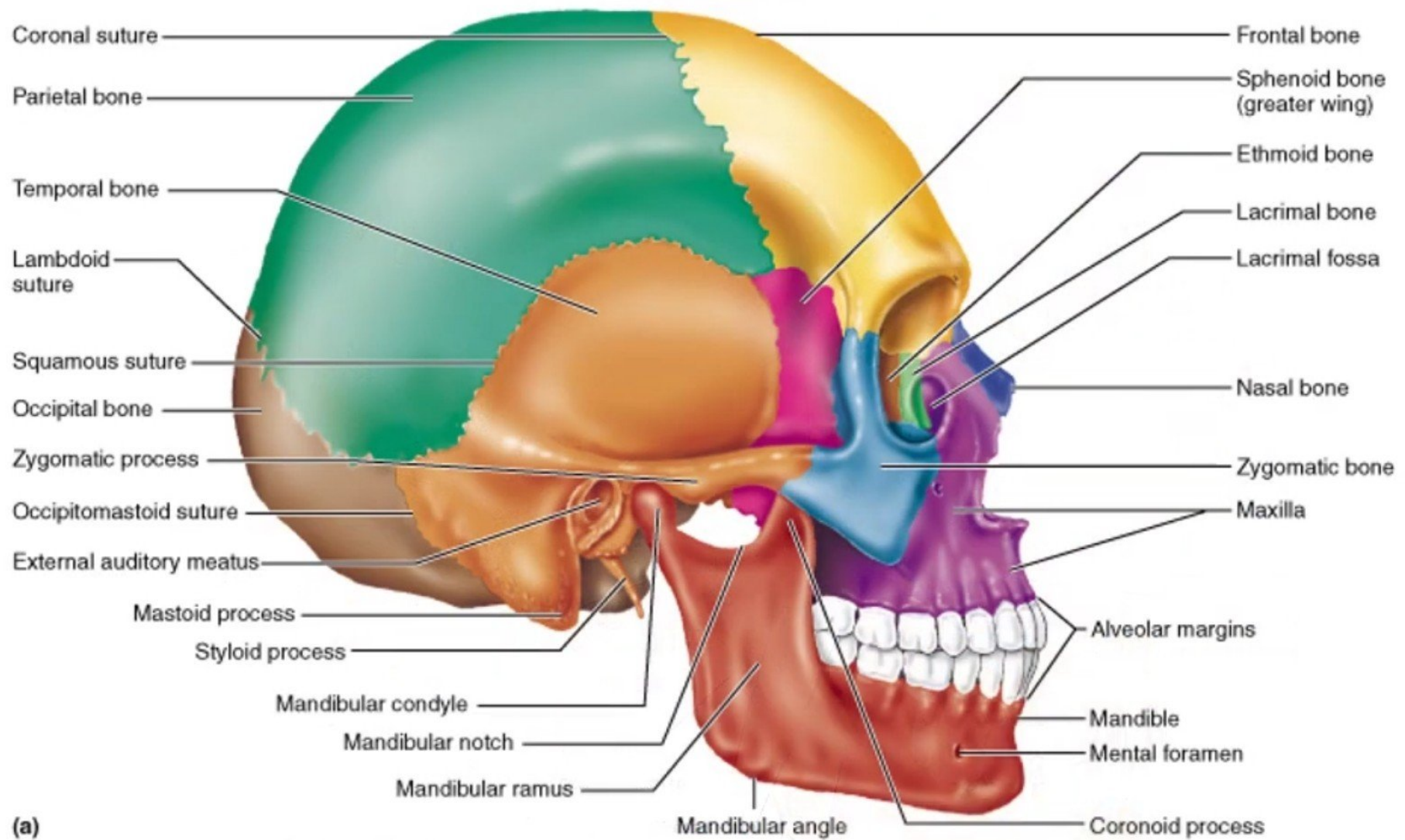






Lateral View

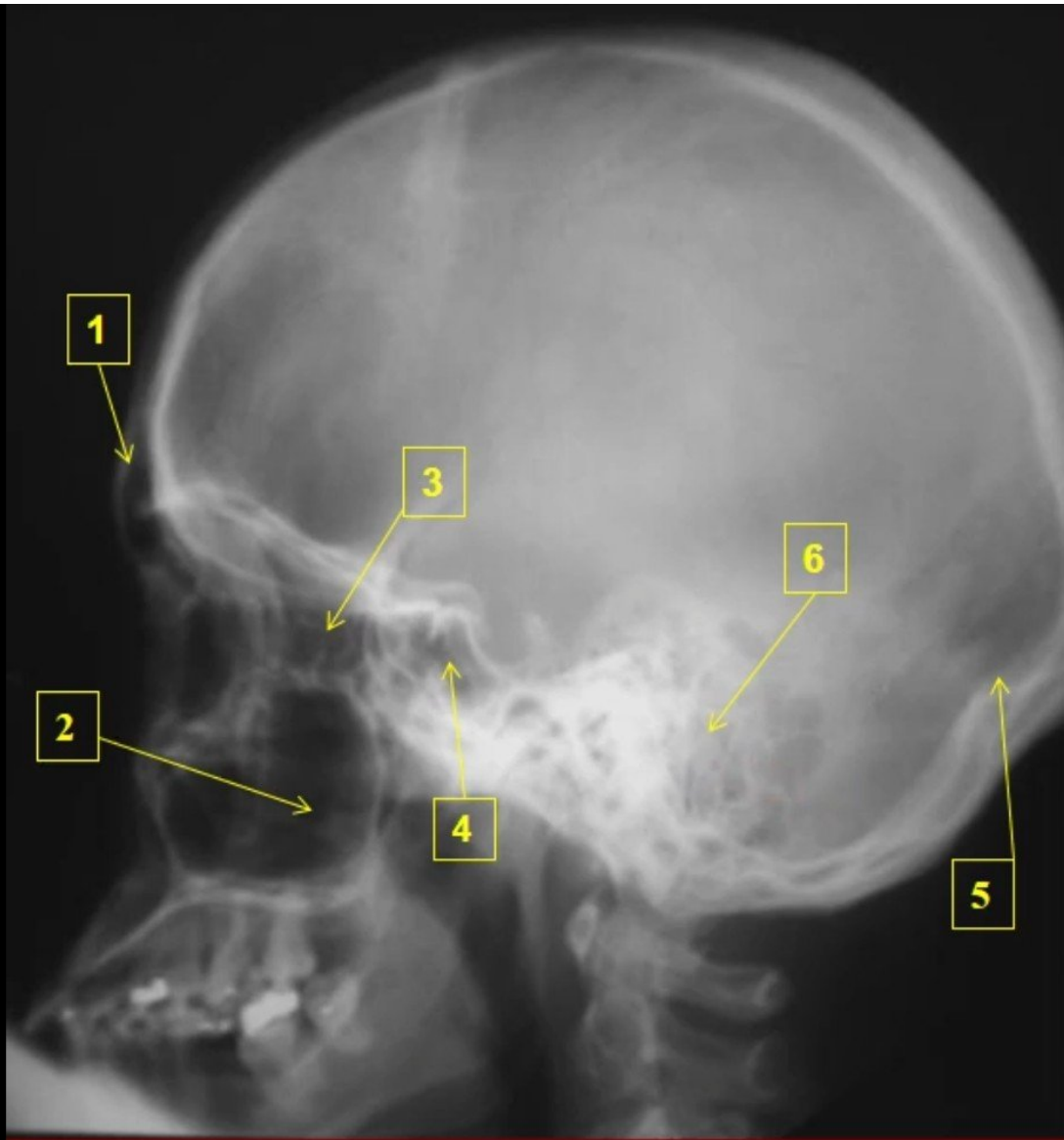




(a)







## Lateral Sinus & Skull

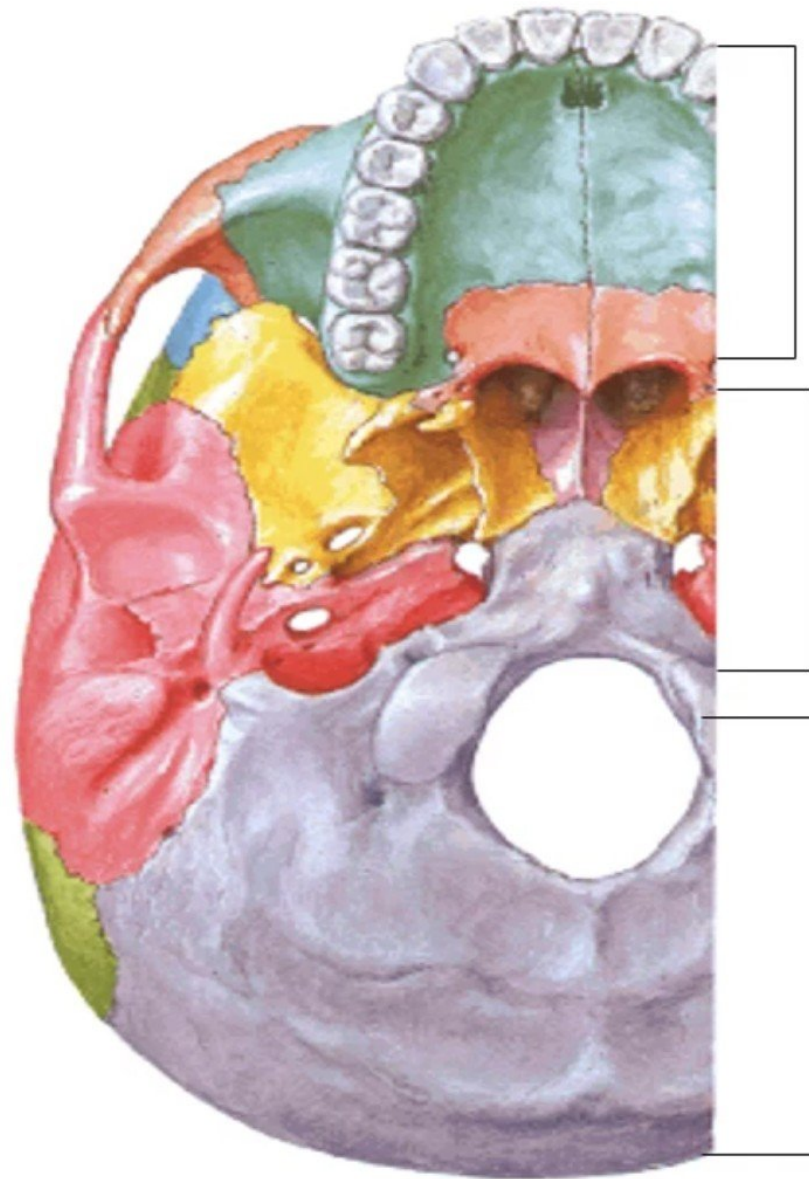
1. Frontal Sinus
2. Maxillary Sinus
3. Ethmoid Sinus
4. Sphenoid Sinus
5. Occipital Bone
6. Mastoid Air Cells





# NORMA BASALIS EXTERNA



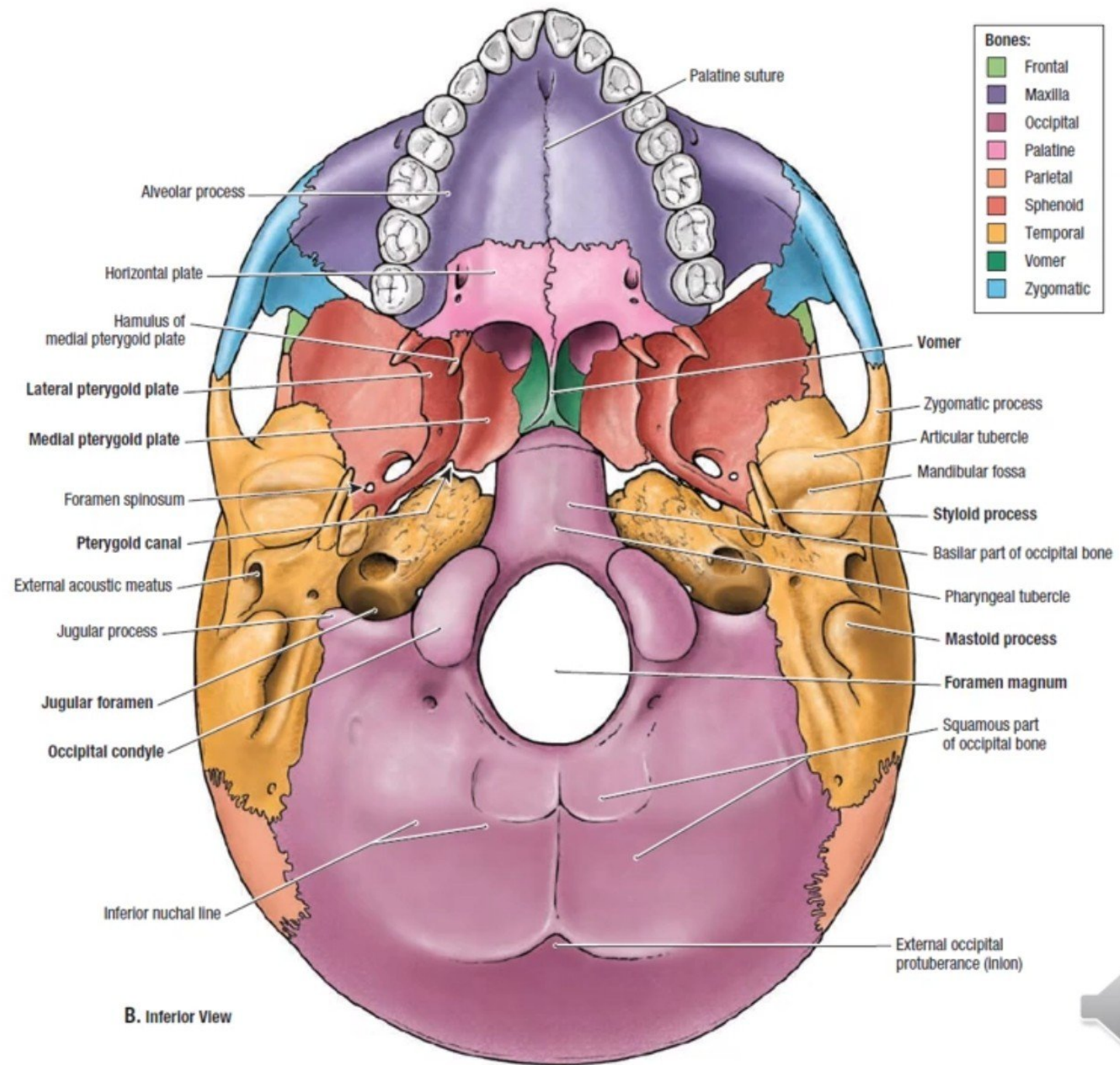


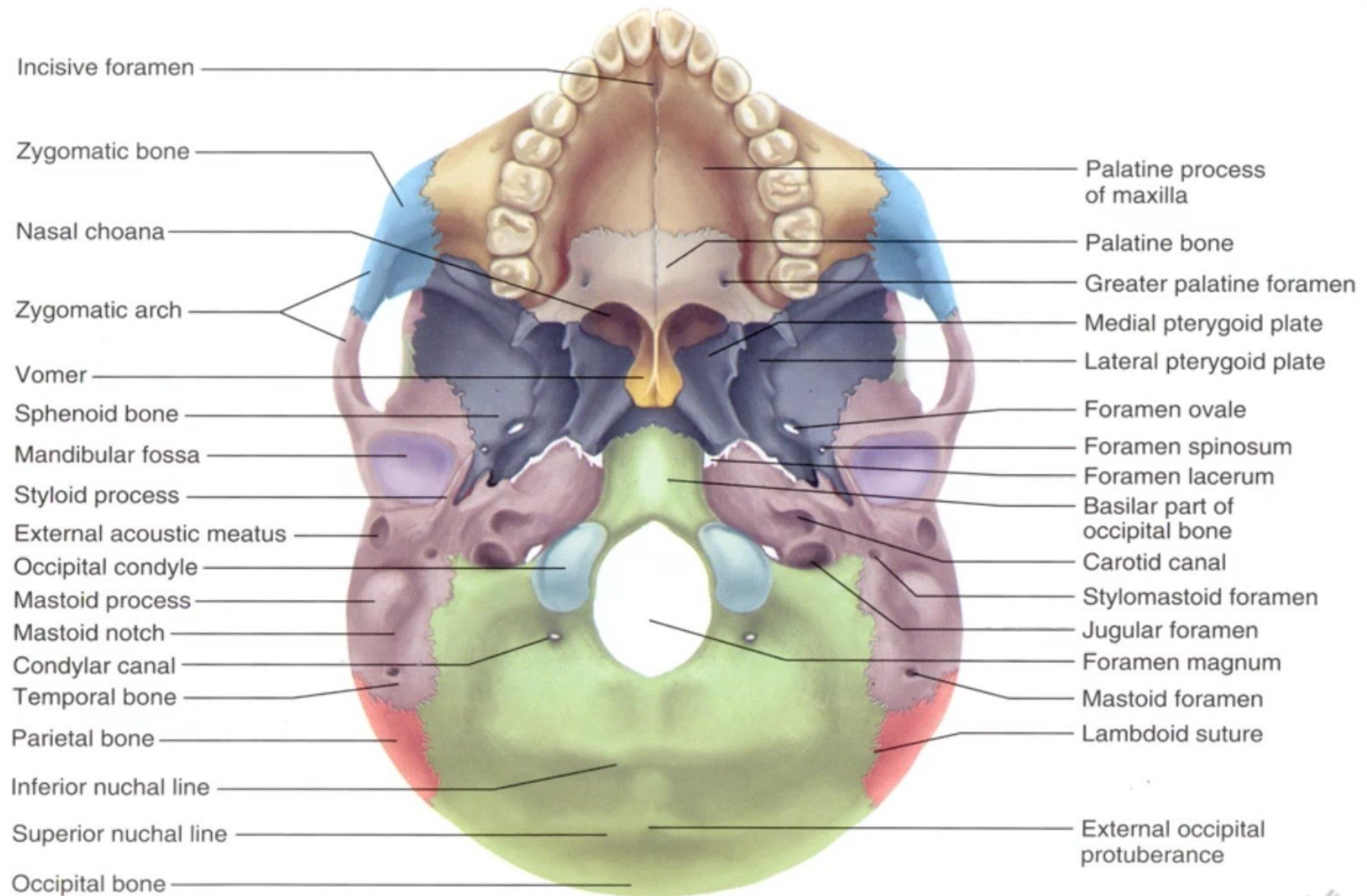
Anterior  
part

Middle  
part

Posterior  
part







# **NORMA BASALIS INTERNA**

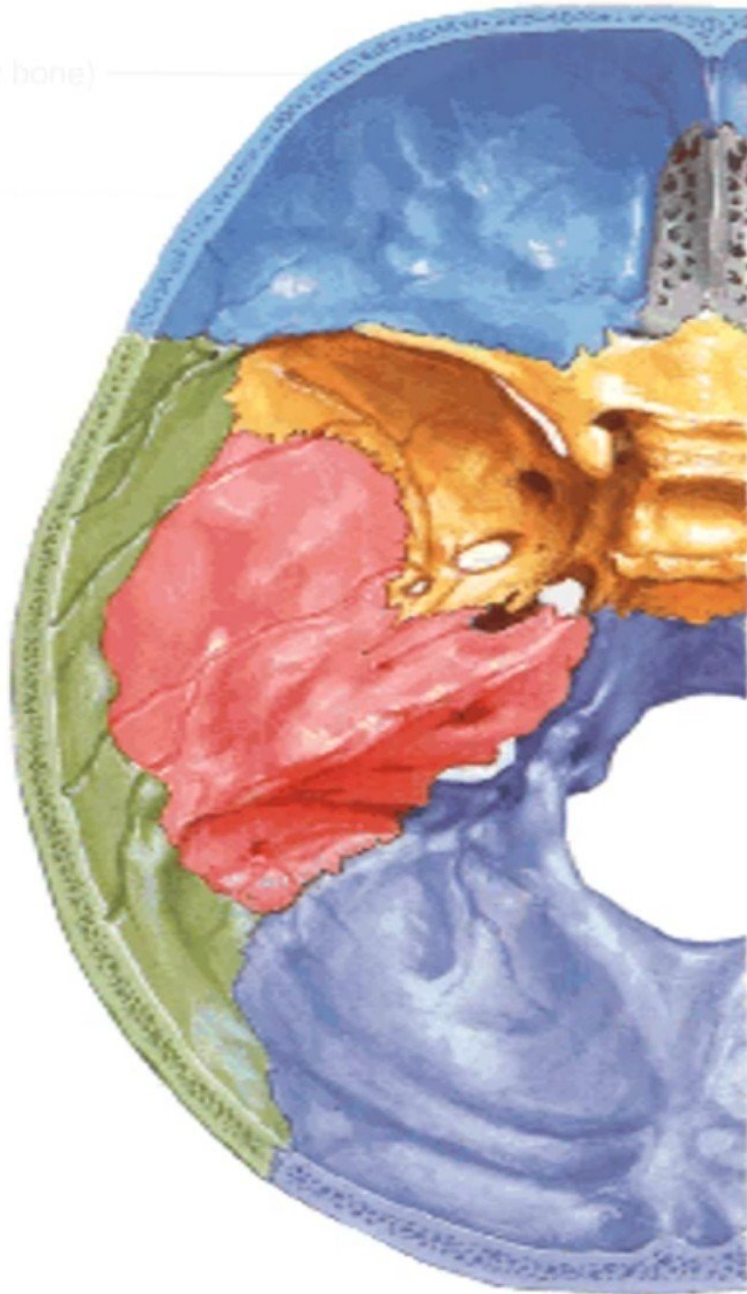


# CRANIAL CAVITY



Diploe (spongy bone)

Foramen



Anterior  
cranial fossa

Middle  
cranial fossa

Posterior  
cranial fossa



Diploe (spongy bone)

Crista galli

Cribriform foramina

Optic foramen

Foramen rotundum

Foramen ovale

Internal acoustic  
meatus

Jugular foramen

Groove for  
venous sinus

Frontal bone

Cribriform plate  
of ethmoid bone

Sphenoid bone

Sella turcica

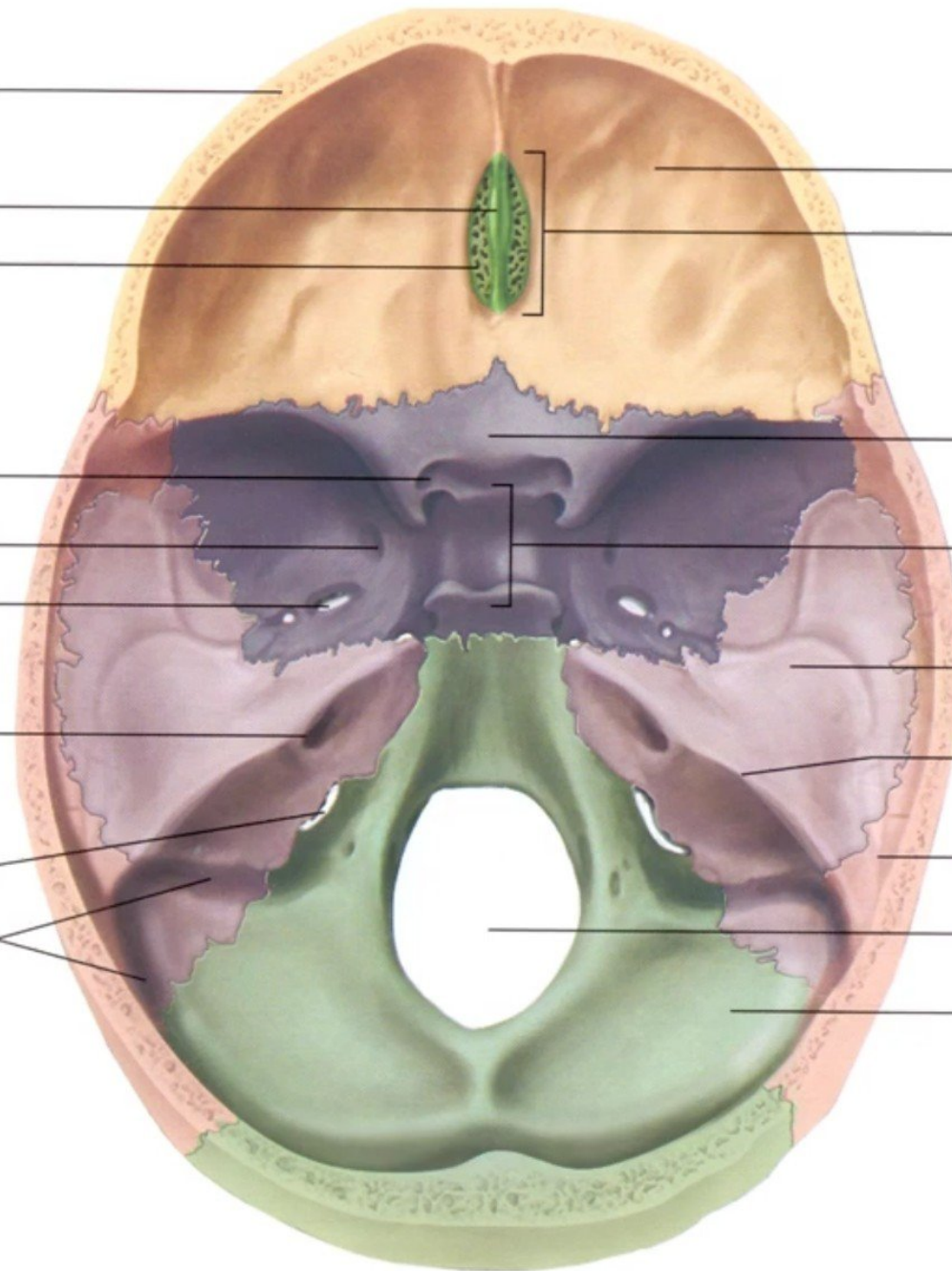
Temporal bone

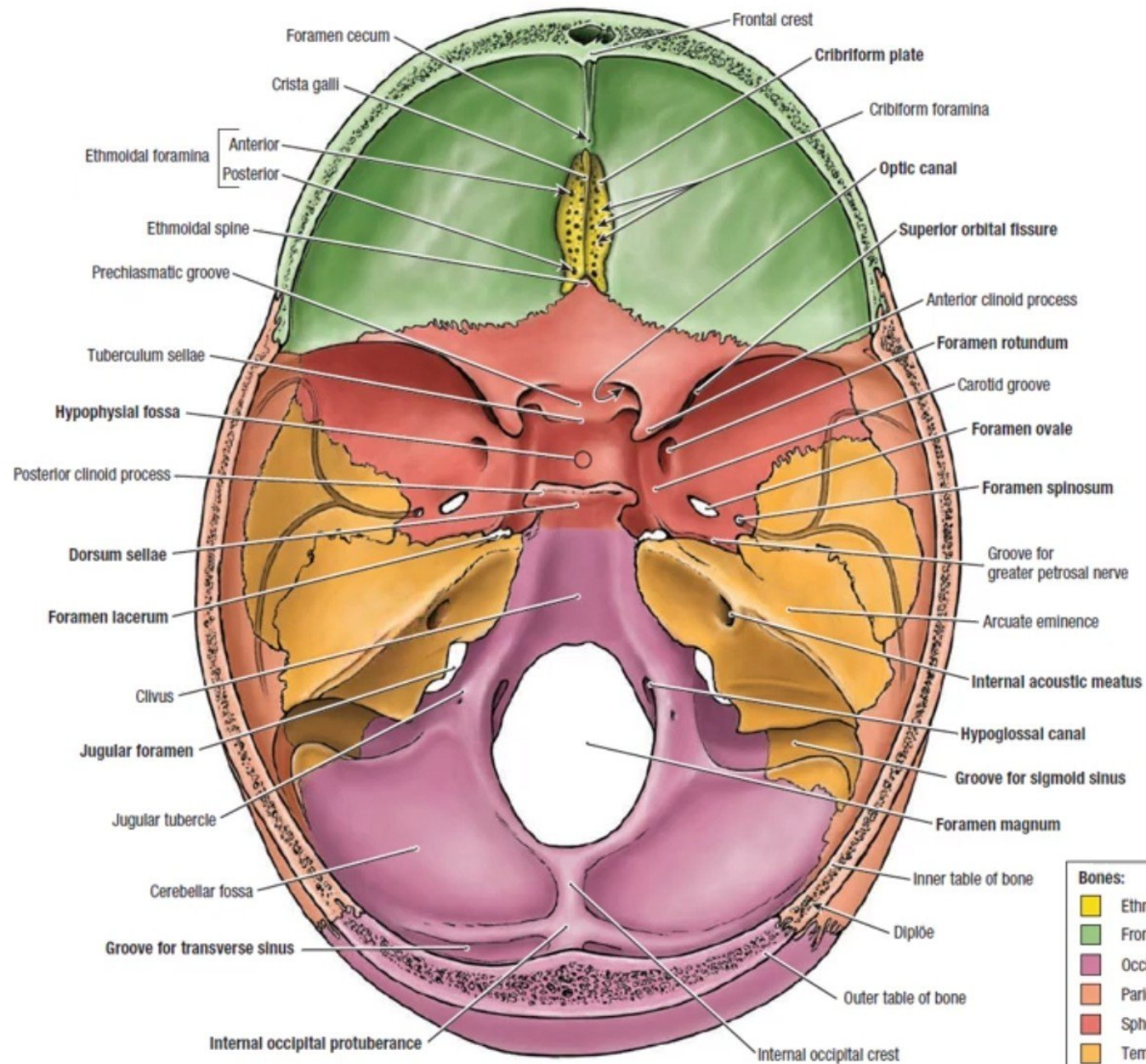
Petrous part of  
temporal bone

Parietal bone

Foramen magnum

Occipital bone

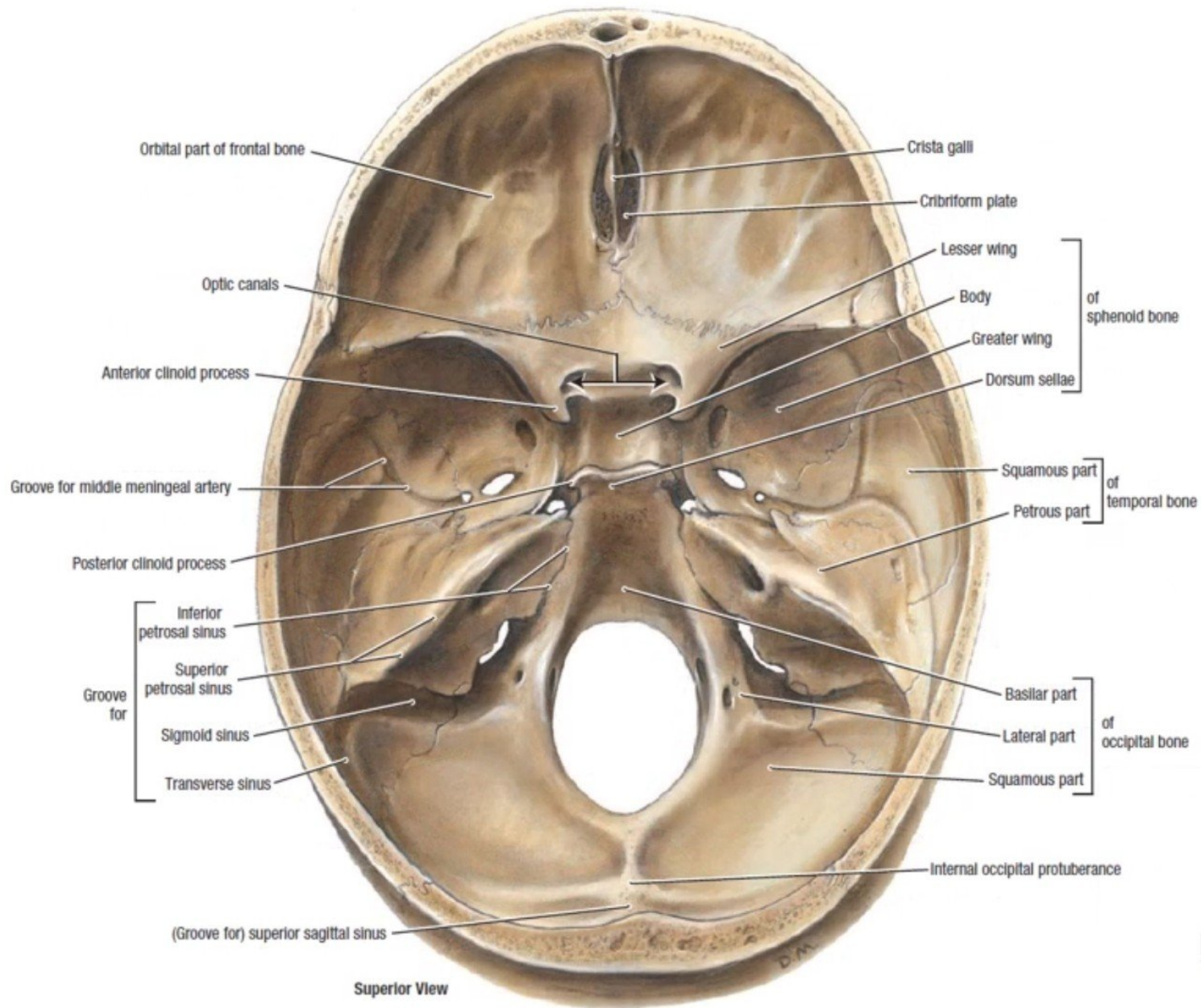




Superior View

| Bones:                                                                                                                     |           |
|----------------------------------------------------------------------------------------------------------------------------|-----------|
| <span style="display:inline-block; width:10px; height:10px; background-color:yellow; border:1px solid black;"></span>      | Ethmoid   |
| <span style="display:inline-block; width:10px; height:10px; background-color:green; border:1px solid black;"></span>       | Frontal   |
| <span style="display:inline-block; width:10px; height:10px; background-color:pink; border:1px solid black;"></span>        | Occipital |
| <span style="display:inline-block; width:10px; height:10px; background-color:orange; border:1px solid black;"></span>      | Parietal  |
| <span style="display:inline-block; width:10px; height:10px; background-color:red; border:1px solid black;"></span>         | Sphenoid  |
| <span style="display:inline-block; width:10px; height:10px; background-color:lightorange; border:1px solid black;"></span> | Temporal  |



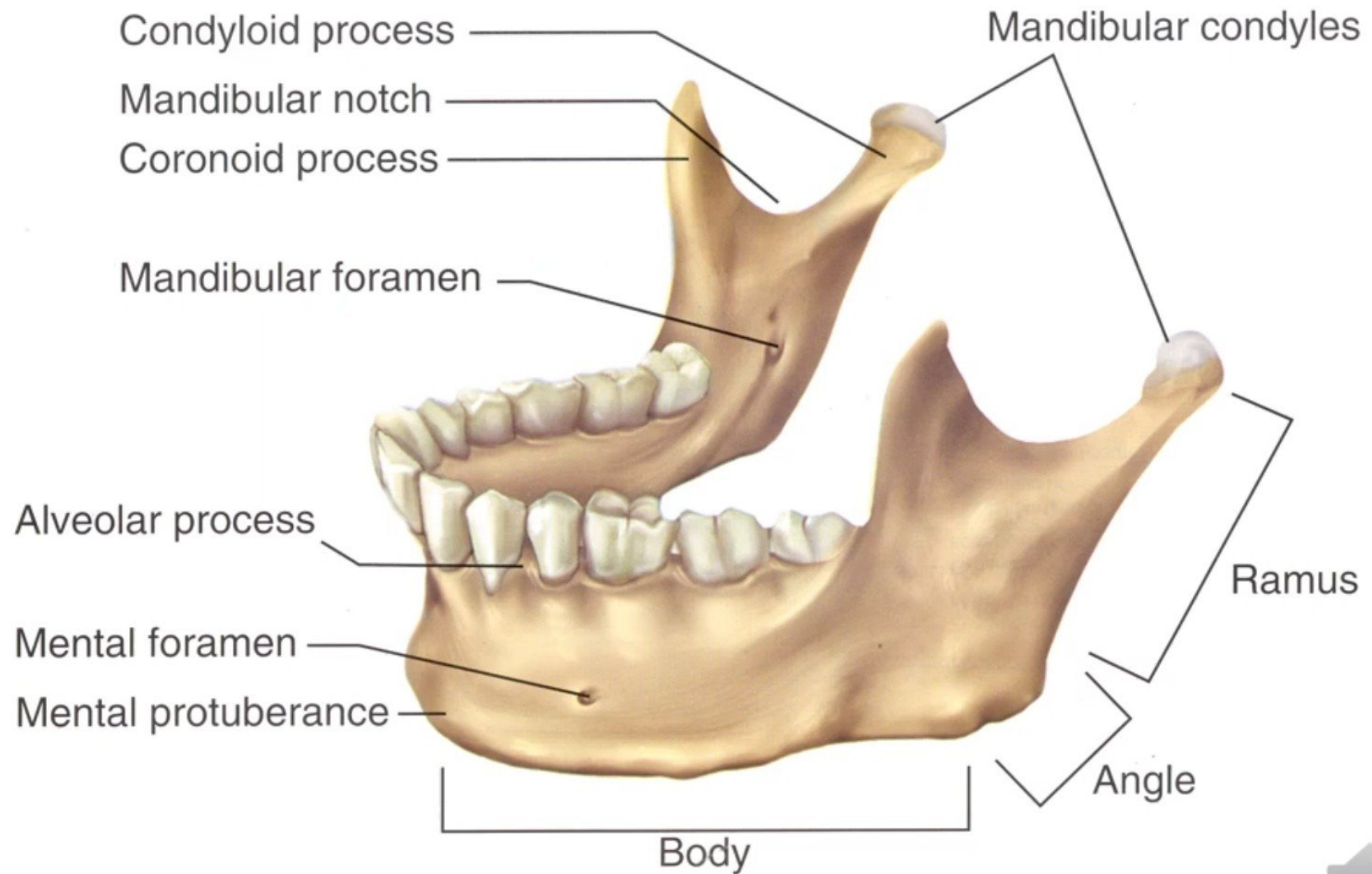


# Skull Base Anatomy Review

## Key Openings

- Foramen spinosum
- Foramen ovale
- Foramen lacerum
- Foramen rotundum
- Foramen magnum
- Jugular foramen
- Superior orbital fissure
- Inferior orbital fissure
- Optic canal
- Hypoglossal canal
- Pterygopalatine fossa









I live inside  
your face.

